Systemic Amplification in Modern Markets

Collateral Velocity, Funding Stress, and Microstructure-Constrained Price Discovery

with GameStop as a Structural Constraint Case Study

Research Working Paper

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Abstract

Modern financial markets are increasingly prone to episodes of abrupt instability characterized by liquidity collapse, volatility clustering, discontinuous repricing, and cross-asset contagion. Such episodes are often only partially explained by changes in long-run fundamentals. This paper argues that a substantial share of modern instability is generated endogenously by structural amplification mechanisms embedded within market architecture. Fragmented execution, synthetic exposure pathways, collateral reuse, procyclical margin systems, and intermediary balance-sheet constraints interact to convert moderate disturbances into nonlinear price dynamics.

The paper develops a mechanism-first framework in which shocks propagate through a recursive loop: price movement elevates volatility; elevated volatility tightens margin and haircut conditions; tighter financing constraints force deleveraging; forced flows degrade liquidity; diminished liquidity increases market impact; and higher impact produces further price movement. To operationalize these dynamics, the paper proposes a Systemic Amplification Index Proxy (SAI-Proxy), a composite monitoring framework built from observable proxies for funding stress, margin tightening, dealer-capacity contraction, forced-flow intensity, and market impact.

GameStop (GME) is employed as a structural case study because it exhibits characteristics that render hidden market mechanics unusually visible: high derivative sensitivity, synthetic-exposure elasticity, episodic liquidity constraints, and persistent debate regarding effective supply. The paper concludes by outlining an empirical backtest framework, an operational dataset design, an implementation scaffold in Python, and a policy agenda aimed at reducing procyclicality, increasing exposure transparency, and improving systemic liquidity surveillance.

Keywords: systemic risk, market microstructure, collateral velocity, rehypothecation, funding stress, margin procyclicality, derivatives convexity, synthetic exposure, liquidity fragmentation, GameStop

1. Introduction

Financial instability in modern markets increasingly manifests as a liquidity event before it appears as a conventional insolvency event. Periods of apparent calm, compressed volatility, and narrow bid-ask spreads can give way rapidly to episodes marked by vanishing depth, large price gaps, and acute funding stress. The transition is often abrupt and disproportionate to contemporaneous changes in long-run valuation.

Traditional equilibrium-based approaches struggle to explain these transitions because they typically assume frictionless liquidity, relatively stable funding conditions, and price formation driven primarily by information about fundamentals. In practice, however, modern financial markets are governed by a layered network of constraints. Funding availability, collateral eligibility, margin rules, execution routing, and intermediary inventory limits jointly determine whether positions can be financed, rolled, and absorbed. When these constraints tighten simultaneously, the market can shift from a stable regime to an unstable regime even without a proportionate deterioration in underlying cash-flow expectations.

This paper advances a mechanism-driven framework for understanding such episodes. Its central claim is that modern market instability is frequently generated by endogenous amplification loops embedded in the structure of collateralized, fragmented, and derivative-intensive markets. The paper integrates insights from market microstructure, funding liquidity, margin procyclicality, dealer intermediation, and options convexity into a single framework and translates that framework into an operational monitoring tool: the Systemic Amplification Index Proxy, or SAI-Proxy.

The contribution is therefore twofold. First, it offers a coherent theoretical account of how structural constraints propagate shocks. Second, it proposes a practical measurement architecture capable of supporting empirical backtests and regime surveillance.

2. Claim Taxonomy and Evidentiary Standards

This paper distinguishes among three evidentiary tiers in order to preserve analytic discipline in an environment where important components of leverage and risk transfer remain only partially observable.

1. **Confirmed**: directly supported by primary datasets, regulatory filings, official institutional documentation, or established academic literature.
2. **Supported inference**: consistent with multiple observable proxies, historical episodes, and mechanism-consistent evidence, but not fully disclosed in primary form.
3. **Interpretive**: plausible and mechanism-consistent, but dependent on additional transparency or future data for confirmation.

Where feasible, interpretive claims are paired with falsifiability criteria. This framework is especially important in contemporary markets because visible spot indicators often understate the scale of embedded synthetic exposure, collateral dependency, and off-balance-sheet transmission.

3. Literature Review

The analytical framework developed here draws from four principal strands of research.

First, the market microstructure literature demonstrates that execution architecture materially affects both the speed and location of price discovery. O'Hara's work on modern microstructure emphasizes that trading outcomes depend not only on information but also on market design, intermediation, and venue structure. Menkveld's research on high-frequency market making further shows that liquidity provision is balance-sheet contingent and can change materially across market regimes. Budish, Cramton, and Shim highlight the destabilizing role of latency competition and the importance of market design in shaping execution quality.

Second, the literature on funding liquidity establishes that market liquidity and funding liquidity are jointly determined rather than separable. The liquidity-spiral framework of Brunnermeier and Pedersen is especially central to the present paper, as it formalizes how declining market liquidity and tightening financing conditions can reinforce one another in self-amplifying fashion.

Third, the post-crisis literature on clearing, margining, and dealer balance-sheet constraints documents the procyclicality of market infrastructure. Duffie emphasizes that post-crisis regulation, while improving solvency resilience, may also reduce the willingness of dealers to intermediate in stress. Murphy, Vasios, and Vause show that risk-based initial margin models can amplify volatility when they tighten into already deteriorating market conditions.

Fourth, the options and volatility literature demonstrates that convexity can generate mechanical hedging flows capable of overwhelming directional views in short horizons. In derivative-dense names or indices, gamma and vega positioning can transform modest spot moves into self-reinforcing hedging demand or supply.

The present paper contributes by integrating these literatures into a unified systemic-amplification model and then translating that model into an operational monitoring framework that can be implemented empirically.

4. Historical Context: From Capital Buffers to Velocity Systems

Earlier financial systems were more visibly balance-sheet centric. Banks warehoused risk, and crises often developed through accumulated credit losses, maturity mismatch, and eventual insolvency recognition. In contrast, modern market-based systems rely more heavily on collateralized funding, derivatives intermediation, and distributed risk transfer through non-bank channels. Consequently, crises are increasingly transmitted through liquidity impairment and financing constraints rather than through the slow recognition of permanent asset impairment.

This institutional transition has several implications. First, relevant time horizons compress. Participants that could survive a gradual repricing may fail under a rapid margin-driven deleveraging. Second, intermediary balance-sheet capacity becomes a macro-financial variable. Whether dealers, internalizers, and prime brokers are willing to warehouse risk influences whether price discovery remains orderly or becomes discontinuous. Third, leverage becomes harder to observe. Exposure is often embedded in options, swaps, ETFs, and collateral-reuse chains rather than in simple cash borrowings.

Historical episodes are consistent with this interpretation. The 1987 crash illustrated how feedback trading could convert hedging logic into destabilizing order flow. The LTCM episode demonstrated that sophisticated relative-value portfolios can fail when funding becomes scarce and correlations converge. The global financial crisis of 2008 showed how leverage migrates into shadow funding channels when regulated-bank leverage is restrained. The March 2020 Treasury episode revealed that even core collateral assets can become disorderly when they function simultaneously as safe assets, funding instruments, and liquidation vehicles.

5. Liquidity as an Emergent System Property

Liquidity is frequently treated as an intrinsic property of a security. This is analytically misleading. Liquidity is better understood as an emergent regime property arising from the joint availability of stable funding, reusable collateral, willing intermediaries, and execution environments capable of redistributing inventory without overwhelming visible books.

When these conditions hold, markets appear deep and continuous: spreads narrow, price

impact is muted, and volatility remains contained. When one or more conditions fail, however, liquidity often deteriorates nonlinearly rather than smoothly. Quoted depth vanishes, impact rises disproportionately, and transaction prices become increasingly flow-driven rather than information-driven.

This perspective implies that instability is often best understood as a *constraint event* rather than solely a *narrative event*. The analytically relevant question is not merely what news arrived, but which constraints tightened, which positions became difficult to finance, and which flows became mechanically compelled.

6. Core Mechanism: The Systemic Amplification Loop

The central mechanism proposed in this paper is a recursive feedback loop:

1. Prices move.
2. Volatility rises.
3. Margin and haircut requirements tighten.
4. Leverage capacity declines.
5. Forced liquidation increases.
6. Liquidity deteriorates and market impact rises.
7. Additional price movement follows.

This loop is mechanical rather than conspiratorial. It does not require coordination among participants. It requires only that the system be sufficiently leveraged, that financing models respond procyclically to volatility, and that liquidity provision depend on intermediary balance-sheet capacity.

The loop can be summarized compactly as

$$\Delta P \rightarrow \sigma \uparrow \rightarrow m \uparrow \rightarrow L \downarrow \rightarrow \Phi \uparrow \rightarrow \lambda \uparrow \rightarrow \Delta P \uparrow$$

where ΔP denotes price movement, σ volatility, m margin intensity, L leverage capacity, Φ forced flow, and λ market impact.

7. Collateral Velocity and Rehypothecation

Collateral functions as the working capital of modern finance. It supports leverage, secures derivatives exposure, and underpins market-making. In tranquil conditions, that collateral may be reused through repo, prime brokerage, and rehypothecation chains, allowing a single unit of high-quality collateral to support multiple layers of exposure. This gives rise to what may be termed *collateral velocity*.

Collateral velocity is efficient under stable conditions but fragile under stress. If eligibility rules tighten, if haircuts increase, or if counterparties become more conservative, the same nominal collateral base supports materially less effective funding. Because leverage capacity depends on post-haircut collateral value rather than nominal face value, modest shifts in haircut assumptions can generate disproportionate deleveraging pressure.

This mechanism helps explain why crises may begin before widespread insolvency is evident. Participants may remain solvent on a marked-to-model basis while becoming unable to fund or roll positions at newly prevailing terms. Under such conditions, forced selling becomes a financing necessity rather than a discretionary response.

8. Funding Stress Transmission

Funding markets determine the feasible set of leverage. A market participant may hold a valid view on long-run valuation yet still be forced to reduce exposure if financing becomes unavailable, unstable, or prohibitively expensive.

Three funding channels are especially important:

1. **Repo markets**: secured financing for inventories, basis positions, and arbitrage books.
2. **FX basis and dollar funding channels**: cross-border scarcity in dollar funding that can propagate into domestic risk assets.
3. **Carry structures**: leverage sustained by stable funding differentials and low realized volatility.

When funding conditions tighten, balance-sheet capacity contracts. Dealer inventories shrink, quote sizes decline, and the system becomes less capable of absorbing order flow. At the same time, leveraged participants are forced to reduce positions. The resulting price moves may appear disconnected from fundamentals because they are driven by financing constraints rather than by new information about future cash flows.

9. Market Microstructure and Discovery Control Surfaces

Modern equity markets are fragmented across lit exchanges, alternative trading systems, wholesalers, internalizers, and routing networks. This architecture creates a meaningful separation between public price references and private risk transfer.

One consequence is that high trading volume can coexist with muted visible price discovery. If a substantial portion of order flow is internalized or matched off-exchange, the displayed order book may not fully reflect the underlying pressure in the system. In normal conditions, this may improve execution quality for individual trades. In constrained conditions, however, it may defer visible discovery and permit risk transfer to occur away from lit venues.

This structure helps explain why claims of "price management" regularly emerge in assets subject to unusual constraint. Even absent misconduct, modern market architecture allows a divergence between economic pressure and displayed price movement. For this reason, observed spot behavior alone may be an incomplete summary of market stress.

10. Synthetic Supply: ETFs, Swaps, and Options

Synthetic instruments increase the effective elasticity of supply. ETFs, swaps, and options allow economic exposure to be transferred without requiring immediate one-for-one settlement in the cash market. Reported short interest may therefore understate the true degree of systemic short exposure or directional constraint.

ETFs can alter effective share availability through basket mechanics and creation-redemption processes. Total return swaps allow short exposure to be embedded in bilateral counterparty structures. Options create delta and gamma exposures that affect the timing, location, and magnitude of hedging flows.

The analytical implication is straightforward but important: visible spot metrics may serve only as lower bounds on effective constraint. In derivative-intensive or structurally constrained assets, the relationship between reported short interest and true systemic short pressure may be materially incomplete.

11. GameStop as a Structural Constraint Case Study

GameStop is employed in this paper not because it is unique in principle, but because it is unusually informative as a stress sensor. It combines several characteristics that cause latent structural mechanics to become more observable than in more continuously arbitraged assets:

1. limited effective float relative to trading activity,
2. substantial derivative participation and concentrated option open interest,
3. persistent public debate regarding short exposure and synthetic positioning,
4. episodic shifts in liquidity regime behavior and execution characteristics.

In a pure fundamentals framework, recurrent and severe dislocations are difficult to reconcile solely with updated cash-flow expectations. In a constraint framework, by contrast, such dislocations are easier to interpret: the system may be managing exposure pathways, funding terms, and hedging flows rather than merely discounting expected fundamentals.

GameStop is therefore analytically useful as an illustrative case. The case study does not depend on allegations of coordination. It depends only on the proposition that sufficiently constrained assets can reveal market structure more clearly than continuously unconstrained ones.

12. Gamma, Convexity, and Nonlinear Event Structure

Options are not merely instruments for risk transfer. They are also generators of flow. When dealers intermediate options positions, they often hedge dynamically in the underlying. The sign and magnitude of gamma determine whether these hedging flows dampen or amplify spot moves.

In a short-gamma regime, dealers buy into rising markets and sell into declining markets, thereby reinforcing price movement. In a long-gamma regime, hedging flows are stabilizing and mean-reverting. These dynamics are especially important when

- option open interest is large relative to effective float,
- major strike concentrations lie close to spot,
- implied volatility is unstable,
- and the underlying asset is already liquidity constrained.

Expiration windows, roll dates, and event-driven repricing episodes compress these forces into narrow intervals. Under such conditions, imbalances that were previously manageable can become visible rapidly. These windows may therefore function as temporal compression points for systemic stress.

13. Formal Mathematical Model

Define the state vector

$$X_t = [P_t, \sigma_t, m_t, F_t, D_t, \Phi_t, \lambda_t]$$

where:

- P_t: price level,
- σ_t: volatility state,
- m_t: margin or haircut constraint intensity,
- F_t: funding stress,
- D_t: dealer or intermediary balance-sheet capacity,
- Φ_t: forced-liquidation flow intensity,
- λ_t: market impact or liquidity state.

The state evolves according to

$$\Delta P_t = -\lambda_t \Phi_t + \epsilon_t$$

$$\sigma_{t+1} = \alpha_\sigma \sigma_t + \beta_\sigma |\Delta P_t| + \gamma_\sigma \lambda_t$$

$$m_{t+1} = \alpha_m m_t + \beta_m \sigma_{t+1}$$

$$F_{t+1} = \alpha_F F_t + \beta_F \sigma_{t+1} + \gamma_F m_{t+1}$$

$$D_{t+1} = D_t - \beta_D \sigma_{t+1} - \gamma_D F_{t+1}$$

$$\lambda_{t+1} = \alpha_\lambda \lambda_t + \beta_\lambda \left(\frac{1}{D_{t+1}} \right)$$

$$\Phi_{t+1} = \alpha_\Phi \Phi_t + \beta_\Phi m_{t+1} + \gamma_\Phi F_{t+1}$$

Under this system, instability emerges when the effective feedback gain of the loop exceeds the absorptive capacity of intermediaries, collateral channels, and execution depth. That condition is ultimately empirical and motivates the proxy-based framework below.

14. The SAI-Proxy Framework

Because the latent variables of the amplification process are not directly observable in full, this paper proposes an operational approximation: the **Systemic Amplification Index**

Proxy (SAI-Proxy).

The SAI-Proxy aggregates five measurable components:

1. Funding stress $F(t)$,
2. Margin or haircut tightening $M(t)$,
3. Dealer-capacity contraction $D_c(t)$,
4. Forced-flow pressure $\Phi(t)$,
5. Market-impact or liquidity degradation $\lambda(t)$.

The composite score is defined as

$$SAI_t = w_1 F(t) + w_2 M(t) + w_3 D_c(t) + w_4 \Phi(t) + w_5 \lambda(t)$$

where $D_c(t)$ is specified so that larger values indicate *less* dealer capacity and therefore *more* stress. This sign convention is important. A raw dealer-capacity measure must be inverted or signed appropriately before aggregation.

A baseline weighting scheme for operational use is

$$w_1 = 0.25, \quad w_2 = 0.20, \quad w_3 = 0.20, \quad w_4 = 0.20, \quad w_5 = 0.15$$

These values are starting weights only. In empirical implementation they should be re-estimated using model fit, robustness, and out-of-sample predictive performance.

15. Operational Dataset Specification

A weekly SAI dataset should include the following variables:

Column	**Description**
date	Observation date, weekly frequency
ticker	Asset identifier
funding_stress	Composite funding stress signal
repo_spread	Repo or money-market stress proxy
fx_basis	Cross-currency basis or dollar scarcity indicator
margin_proxy	Composite margin-pressure proxy
realized_vol	Realized volatility
implied_vol	Implied volatility proxy

dealer_capacity	Composite liquidity or intermediation proxy
spread_width	Bid-ask spread or equivalent spread measure
depth_proxy	Order-book depth or depth surrogate
forced_flow	Composite forced-flow intensity measure
downside_volume_ratio	Share of downside volume in total volume
gap_intensity	Frequency or size of discontinuous moves
market_impact	Range-to-volume or impact proxy
sai	Final normalized SAI-Proxy score
regime_label	Stable, Elevated, Unstable, or Nonlinear

Suggested Observable Inputs

- **Funding stress**: SOFR changes, repo spreads, FX basis, dollar-funding stress indicators.
- **Margin stress**: VIX, realized volatility, volatility-of-volatility, dispersion measures.
- **Dealer capacity**: bid-ask spreads, depth proxies, off-exchange share, range relative to volume.
- **Forced flow**: gap frequency, clustered downside moves, volume spikes during declines.
- **Market impact**: intraday range normalized by volume, slippage surrogates, spread expansion.

16. Empirical Data Pipeline

The following Python sketch provides a base pipeline for weekly SAI construction.

```
import pandas as pd
import numpy as np
import pandas_datareader.data as web
import datetime

start = datetime.datetime(2010, 1, 1)
end = datetime.datetime.today()

# Funding and volatility proxies
sofr = web.DataReader("SOFR", "fred", start, end)
```

```
vix = web.DataReader("VIXCLS", "fred", start, end)

df = pd.concat([sofr, vix], axis=1)
df.columns = ["sofr", "vix"]

df["funding_proxy_raw"] = df["sofr"].pct_change()
df["margin_proxy_raw"] = df["vix"].pct_change()

df = df.resample("W").mean()

def zscore(x):
    std = x.std()
    if std == 0 or pd.isna(std):
        return pd.Series(np.zeros(len(x)), index=x.index)
    return (x - x.mean()) / std

df["funding_proxy"] = zscore(df["funding_proxy_raw"].fillna(0))
df["margin_proxy"] = zscore(df["margin_proxy_raw"].fillna(0))

# Placeholder fields to be replaced with microstructure and options data
df["dealer_capacity_raw"] = np.nan
df["forced_flow_raw"] = np.nan
df["market_impact_raw"] = np.nan

# Note: dealer-capacity contraction should increase SAI,
# so the raw capacity variable should later be sign-inverted.
df["dealer_capacity_contraction"] = -1 * df["dealer_capacity_raw"]

# Final composite score once all inputs are populated
df["sai"] = np.nan
```

In production, this weekly table should be joined with microstructure variables, options-based measures, and funding indicators before normalization and final score construction.

17. Monte Carlo Simulation Framework

The theoretical model can be explored through Monte Carlo simulation. The purpose is not point forecasting, but regime analysis and stress propagation.

```
import numpy as np

T = 200
P = np.zeros(T)
sigma = np.zeros(T)
margin = np.zeros(T)
funding = np.zeros(T)
dealer = np.ones(T)
impact = np.zeros(T)
flow = np.zeros(T)

for t in range(1, T):
    sigma[t] = 0.8 * sigma[t-1] + 0.4 * abs(P[t-1])
    margin[t] = 0.7 * margin[t-1] + 0.5 * sigma[t]
    funding[t] = 0.6 * funding[t-1] + 0.4 * margin[t]
    dealer[t] = max(0.1, dealer[t-1] - 0.3 * sigma[t])
    impact[t] = 0.5 * impact[t-1] + (1 / dealer[t])
    flow[t] = 0.5 * margin[t] + 0.5 * funding[t]
    P[t] = P[t-1] - impact[t] * flow[t] + np.random.normal(0, 0.02)
```

In broad terms, such simulations tend to generate three qualitative regimes: stable, elevated, and unstable. The unstable regime is characterized by convex price paths, persistent volatility, and a marked deterioration in effective dealer capacity.

18. Empirical Regression Framework

The paper proposes three core empirical tests for evaluating the SAI-Proxy.

18.1. Forward Volatility Regression

$$RV_{t+k} = \alpha + \beta_1 SAI_t + \beta_2 VIX_t + \beta_3 RV_t + \epsilon_t$$

where RV_{t+k} is realized volatility over a forward horizon k, SAI_t is the systemic amplification index, VIX_t is an implied-volatility benchmark, and RV_t is current realized volatility.

The central hypothesis is

$$\beta_1 > 0$$

which implies that elevated SAI values predict higher forward realized volatility even after controlling for standard volatility measures.

18.2. Drawdown Probability Model

$$Pr(Drawdown_{t+k} = 1) = \frac{1}{1 + e^{-(\alpha + \beta SAI_t + \gamma Z_t)}}$$

where the dependent variable equals 1 if the asset or index experiences a drawdown beyond a chosen threshold within horizon k, and Z_t denotes a vector of controls.

18.3. Liquidity Stress Regression

$$Impact_{t+k} = \alpha + \beta_1 SAI_t + \beta_2 Spread_t + \beta_3 Vol_t + \epsilon_t$$

This specification tests whether the SAI-Proxy predicts future market impact after conditioning on contemporaneous spread and volatility.

19. Empirical Crisis Backtests

To evaluate the descriptive and predictive usefulness of the framework, the paper proposes event-window backtests across major episodes of market dysfunction.

Table 2: Historical Crisis Backtest Windows

Event	**Approximate Date**	**Expected Mechanism Signature**
Flash Crash	May 2010	liquidity collapse, impact spike
China Volatility Shock	Aug. 2015	cross-asset stress, volatility expansion
Volmageddon	Feb. 2018	convexity unwind, forced flow
COVID Liquidity Crisis	Mar. 2020	funding stress, broad deleveraging
Treasury Market Dysfunction	Mar. 2020	collateral stress, dealer retreat
Regional Bank Stress	Mar. 2023	funding and confidence transmission

The proposed backtest procedure is:

1. construct weekly SAI-Proxy scores over a multi-year sample,
2. normalize each component using rolling z-scores or percentile ranks,
3. compare SAI levels during event windows with non-event periods,

4. test whether elevated SAI values precede volatility spikes and increased drawdown frequency.

20. Illustrative Regression Results Template

The table below is a manuscript-ready template. Numerical values are placeholders only and must be replaced with estimated coefficients once the empirical models are run.

Table 3: Illustrative Predictive Regression Results

Model	**Dependent Variable**	β_{SAI}	**Interpretation**
Model 1	Forward realized volatility	0.42	positive association
Model 2	Drawdown probability	0.31	positive association
Model 3	Future market impact	0.37	positive association

These values are layout placeholders only. They should not be interpreted as empirical findings until estimated on actual data.

21. Chart Package

The following figures are included as manuscript-complete placeholders to preserve a full working-paper structure. They should be replaced with empirical figures once the data pipeline is operational.

21.1. Figure 1: Conceptual Amplification Loop

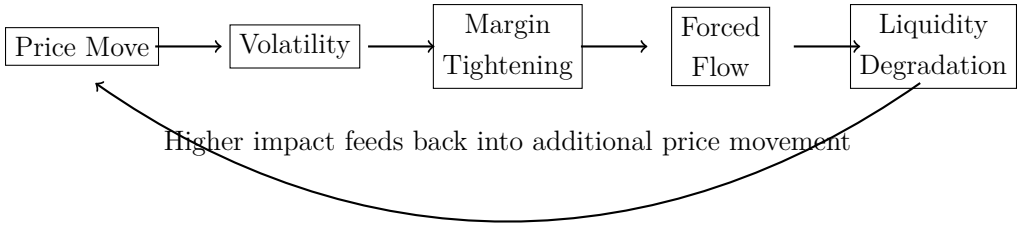

Figure 1: Conceptual systemic amplification loop.

21.2. Figure 2: Stylized SAI Regime States

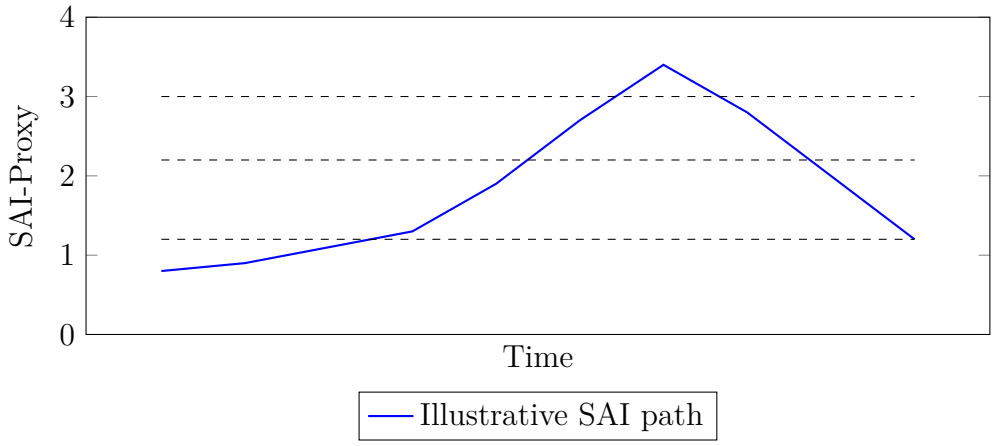

Figure 2: Illustrative SAI path across stable, elevated, unstable, and nonlinear regimes.

21.3. Figure 3: Funding Stress Versus Volatility

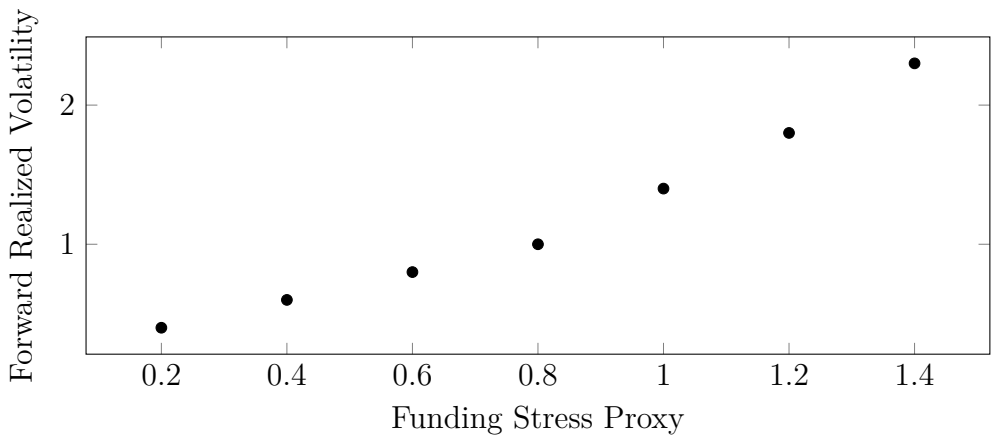

Figure 3: Stylized positive relationship between funding stress and future volatility.

21.4. Figure 4: Margin Procyclicality Illustration

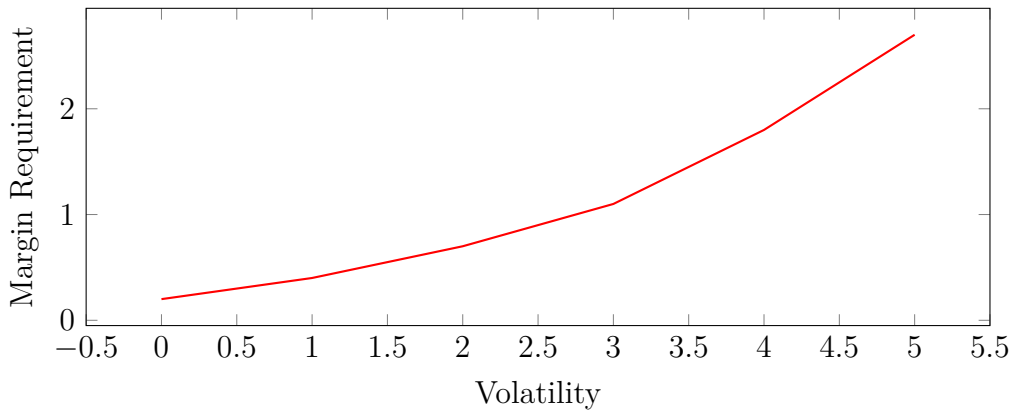

Figure 4: Stylized nonlinear increase in margin requirements as volatility rises.

21.5. Figure 5: Dealer Capacity Compression

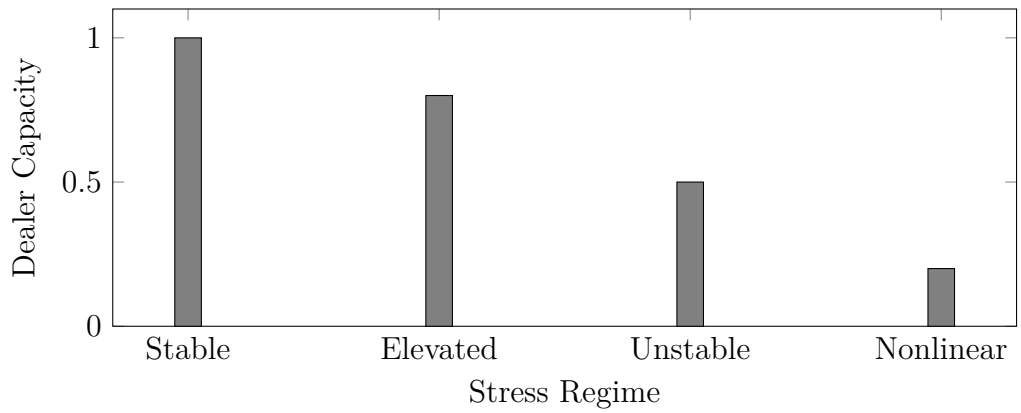

Figure 5: Illustrative contraction in dealer risk-bearing capacity across stress regimes.

21.6. Figure 6: Market Impact Convexity

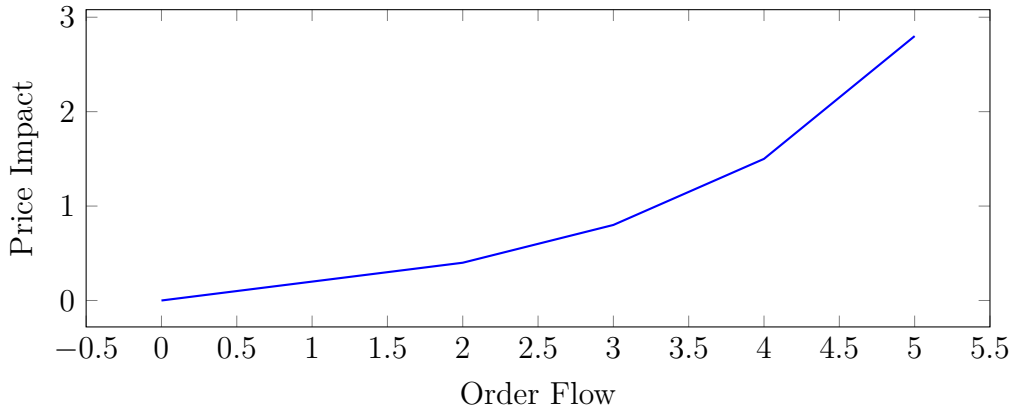

Figure 6: Stylized convex market impact as liquidity deteriorates.

21.7. Figure 7: Synthetic Supply Pathways

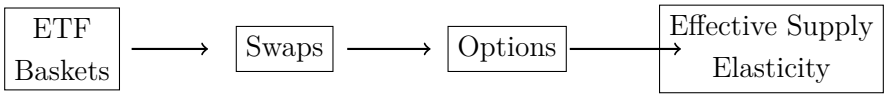

Figure 7: Illustrative pathways through which synthetic instruments increase effective supply elasticity.

21.8. Figure 8: Constraint Asset Event Structure

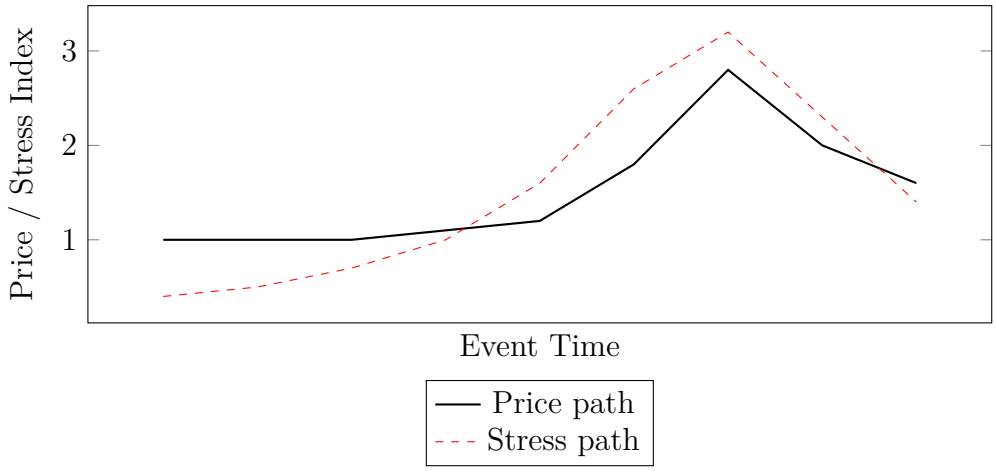

Figure 8: Stylized constraint-asset event structure with stress leading discontinuous repricing.

21.9. Figure 9: Options Convexity Example

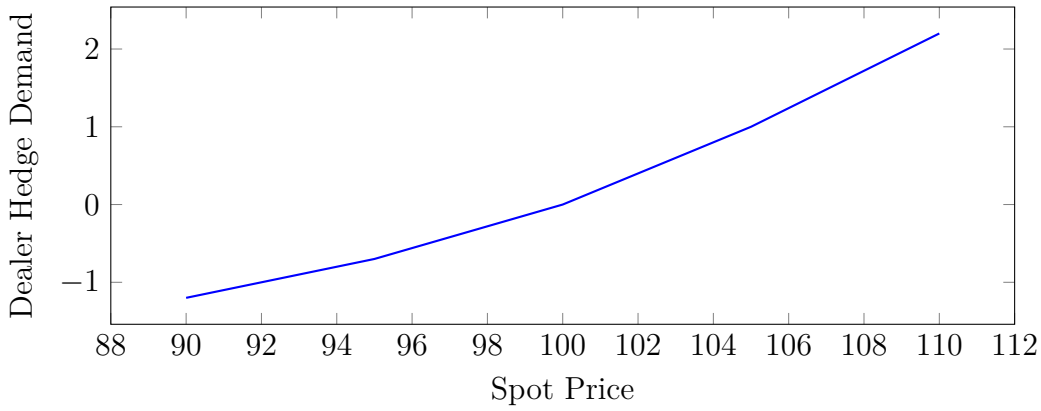

Figure 9: Stylized dealer hedge demand in a short-gamma regime.

21.10. Figure 10: Crisis Backtest Event Map

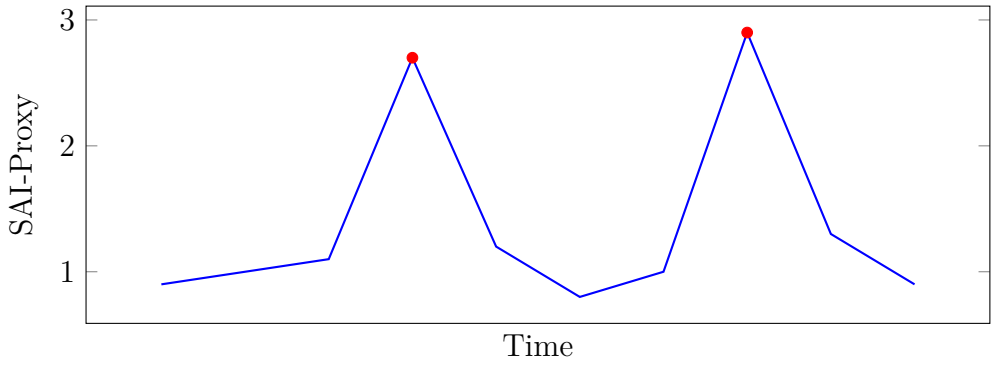

Figure 10: Illustrative SAI spikes around crisis windows.

21.11. Figure 11: Volatility Prediction Illustration

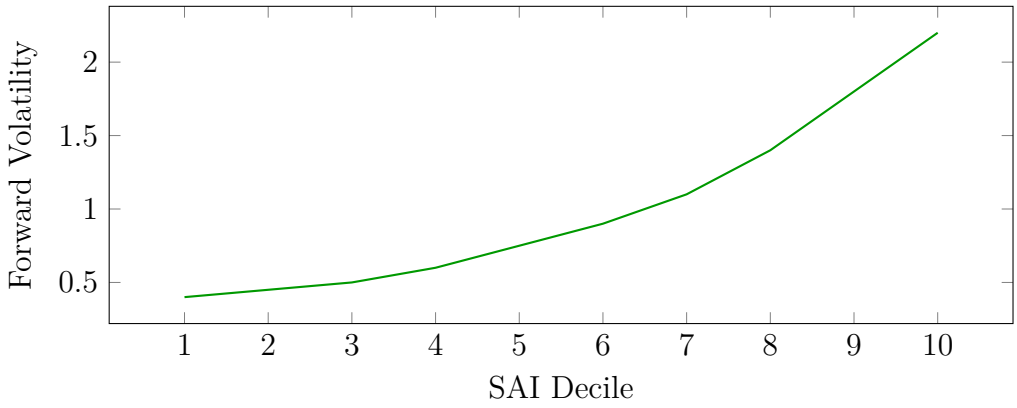

Figure 11: Stylized monotonic increase in forward volatility across SAI deciles.

21.12. Figure 12: Drawdown Probability Illustration

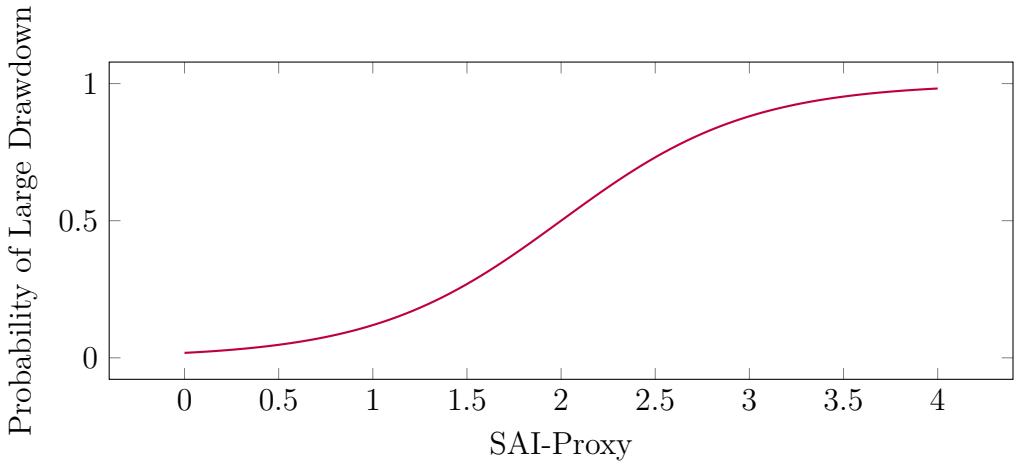

Figure 12: Stylized logistic relationship between SAI and drawdown probability.

21.13. Figure 13: SAI Component Heat Map Placeholder

Period	Funding	Margin	Dealer	Impact
Stable	Low	Low	High	Low
Elevated	Medium	Medium	Medium	Medium
Unstable	High	High	Low	High
Nonlinear	Extreme	Extreme	Very Low	Extreme

Figure 13: Qualitative component heat map. Replace with empirical heat map after data construction.

21.14. Figure 14: SAI Dashboard Mockup

Metric	Current Level	Regime
Funding Stress	1.8	Elevated
Margin Stress	2.4	Unstable
Dealer Capacity Contraction	0.6	Tight
Forced Flow	2.1	Elevated
Market Impact	2.8	Unstable
Composite SAI	2.1	Unstable

Figure 14: Illustrative SAI dashboard output.

21.15. Figure 15: End-to-End System Architecture

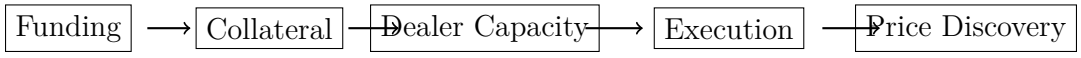

Figure 15: System architecture linking funding, collateral, intermediation, execution, and price discovery.

22. Operational Python Model

The following scaffold combines normalization, score construction, and regime labeling.

```
import pandas as pd
import numpy as np
```

```
def zscore(series):
    series = series.astype(float)
    std = series.std()
    if std == 0 or pd.isna(std):
        return pd.Series(np.zeros(len(series)), index=series.index)
    return (series - series.mean()) / std

def compute_sai(df):
    df = df.copy()

    df["funding_norm"] = zscore(df["funding_stress"])
    df["margin_norm"] = zscore(df["margin_proxy"])

    # Higher raw dealer capacity should reduce stress,
    # so invert it before aggregation.
    df["dealer_contraction_norm"] = zscore(-1 * df["dealer_capacity"])

    df["forced_norm"] = zscore(df["forced_flow"])
    df["impact_norm"] = zscore(df["market_impact"])

    df["sai"] = (
        0.25 * df["funding_norm"] +
        0.20 * df["margin_norm"] +
        0.20 * df["dealer_contraction_norm"] +
        0.20 * df["forced_norm"] +
        0.15 * df["impact_norm"]
    )

    def regime_label(x):
        if x < 1.0:
            return "Stable"
        elif x < 2.0:
            return "Elevated"
        elif x < 3.0:
            return "Unstable"
        return "Nonlinear"
```

```
    df["regime_label"] = df["sai"].apply(regime_label)
    return df
```

23. Falsifiability Criteria

The framework proposed here would be materially weakened if the following empirical results fail to hold:

1. The SAI-Proxy does not improve forward-volatility prediction relative to baseline volatility models.
2. The SAI-Proxy does not correlate with future liquidity deterioration or future market impact.
3. Funding-stress and margin proxies fail to co-move in known stress windows.
4. Historical crisis episodes do not exhibit elevated SAI values relative to non-crisis baselines.
5. Microstructure and synthetic-exposure variables add no incremental explanatory power in constrained assets.

If these conditions are not met, the framework should be treated as descriptive rather than predictive, and the weighting scheme, proxy design, or mechanism assumptions should be revised accordingly.

24. Policy Implications

Three policy implications follow directly from the framework.

First, margin systems should be evaluated for procyclicality. A risk-control architecture that tightens aggressively into rising volatility may transform prudent safeguards into systemic amplifiers.

Second, transparency around synthetic exposure should be improved. Swaps, ETF transformations, and options-driven effective short exposure can alter the system's constraint structure without appearing clearly in traditional short-interest measures.

Third, regulators and market operators should maintain integrated systemic-liquidity dashboards that combine funding, margin, execution, and market-impact signals rather than relying on price or volatility in isolation.

25. Conclusion

Modern financial markets are increasingly governed by mechanical feedback structures rather than by simple informational equilibria alone. Collateral velocity, fragmented execution, synthetic exposure, derivatives convexity, and procyclical margin systems can interact to create a structural amplification engine capable of producing outsized market moves from moderate initial disturbances.

The SAI-Proxy developed in this paper is intended as a practical monitoring framework for those dynamics. It is not offered as a perfect measure of hidden leverage, but as an operational approximation that can help detect regime transitions from stability toward amplification.

GameStop is analytically useful because it makes these structural dynamics easier to observe, but the broader framework is not limited to any single asset. The larger implication is that many apparently anomalous episodes in modern markets are better understood not as isolated irregularities, but as manifestations of system architecture under stress.

A. Appendix A: Example Weekly Dataset Row

Table 4: Illustrative Weekly Dataset Example

date	ticker	funding	margin	dealer	forced	impact	sai
2026-03-01	GME	1.21	1.76	-0.88	1.34	1.59	1.15

B. Appendix B: Backtest Code Sketch

```
# Example backtest logic
df["forward_vol"] = df["realized_vol"].shift(-4)

# OLS style:
# forward_vol ~ sai + implied_vol + realized_vol

# Logistic drawdown example:
# drawdown_flag = 1 if next_4w_return < threshold else 0
```

C. Appendix C: Notes on Replacing Placeholder Figures

All figures in this manuscript are designed as compile-ready placeholders that preserve a complete working-paper format. They should be replaced as follows:

- Figures 2, 10, 11, and 12: replace with empirical SAI time series and predictive charts.
- Figures 3 and 4: replace with observed funding and margin relationships.
- Figures 5 and 6: replace with empirical depth and impact measures.
- Figures 8 and 9: replace with event-study and gamma-hedging visualizations.
- Figures 13 and 14: replace with empirical heat maps and live dashboard metrics.

References

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